Predicting Star Contracts in the NBA

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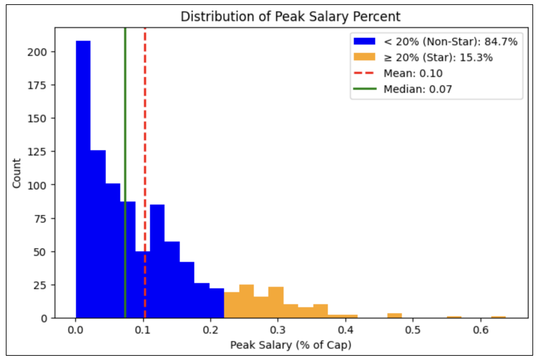
Introduction

In the NBA, teams invest heavily in young players, yet projection of long-term contract value is less studied. While performance and salary have both been analyzed together, less is known about how early-career production and development translate into future earnings. This project asks: **which early-career performance factors lead to a "star-level" contract during a player's career?**

Data Collection

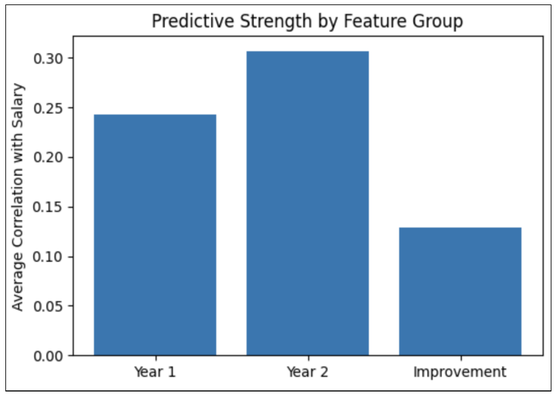
This dataset utilizes comprehensive player information from 1991 to 2015, focusing on 924 individuals who played at least two professional seasons. By leveraging Gabriel Pastorello's **BRScraper** tool, advanced metrics and per-game averages, like True Shooting percentage to Value Over Replacement Player, were extracted from Basketball-Reference for each player's first two seasons. These statistics were then integrated with contract data meticulously scraped from individual team-season pages to align player salaries with their calculated maximum potential earnings.

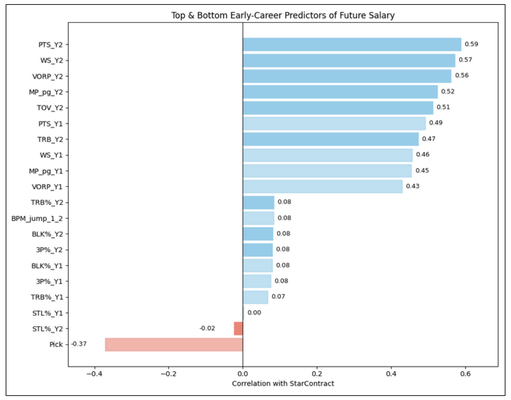
Results



- Only **15.3%** of players in this study reached the "Star" designation (earning ≥ 20% of the salary cap), while the vast majority (84.7%) remained in the "Non-Star" bracket.
- The median peak salary is just **7%** of the cap; the distribution falls off sharply, with only a tiny handful of elite players ever crossing the 40% or 50% threshold due to changes in contract rules in the NBA.

- Year 2 statistics are significantly more predictive of future salary than Year 1 stats, suggesting that a player's sophomore performance is a much more reliable indicator of their long-term value.
- Fans love to talk about "most improved" players. However, actual statistical improvement between Year 1 and Year 2 has a much weaker correlation with future salary compared to their raw performance levels.





- Scoring (PTS_Y2) and comprehensive metrics like Win Shares (WS_Y2) and VORP show the strongest positive correlations (0.56–0.59) with landing a "Star Contract."
- The "Pick" variable has a strong negative correlation (-0.37), confirming that lower-numbered picks (top of the draft) are much more likely to earn high-end salaries than late-round picks.

Conclusions

- Early-career performance explains a portion of long-term salary outcomes, but not all variation. Non-statistical factors, such as team context, injuries, and contract structures, also influence earnings.
- Players who demonstrate the ability to consistently produce early in their careers are more likely to reach star-level contracts than those relying primarily on efficiency or short-term improvement.
- In particular, sustained output across multiple statistical categories appears to signal long-term value more effectively than isolated metrics.
- These findings suggest that teams prioritize reliability and role growth when evaluating talent, offering insight into how early performance is translated into financial decision-making in professional basketball.

Model Performance

- PCA: R^2 of .38
- OLS: R^2 of .48
- Lasso/Ridge: R^2 of .47

Key Takeaways

- Individual metrics stronger than improvement
- Production & opportunity matter most
- Most models not particularly accurate, signifying that further information is needed to consistently predict star-level contracts with accuracy

Influential Variables

- Points, WS, VORP, minutes most strongly correlated with future star-level contracts
- Draft position also influential